Machine Learning & Neural Networks for Neuroscience

Homework 1: Regression

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Contribution of each team member

Joint work. We worked together on selecting neurons for analysis, interpreting the fitted models and their parameters, discussing the results, and answering the theoretical questions.

Nandana Dileep (computer science background). Nandana mainly focused on organizing and cleaning up the Python code, making the plots consistent, and uploading the final code to GitHub. She also helped with the mathematical and machine learning aspects of the assignment, including model fitting and optimization.

Daniil Miloserdov (neuroscience and data science background). Danii mainly focused on the neuroscience aspects of the assignment, including understanding the dataset structure, explaining the experimental setup, and discussing the biological meaning of the model parameters. He also prepared and formatted the final PDF report.

Use of AI tools. Figures 4 and 5 are schematic illustrations generated with ChatGPT and are not based on the recorded data. AI was also used to format the text; AI was not used to interpret the results obtained.

1 From raw spikes to a tuning curve

Data processing

We analyzed a recording session from the macaque visual-cortex dataset, using the V1 spike-count array from session `106r002p70`. The array has shape $(3200, 1280, 88)$, corresponding to trials, time bins, and V1 neurons, respectively. Each trial contains 1280 one-millisecond bins, covering the 1280 ms stimulus presentation period.

To measure the response of neuron n in trial i, we summed its spike counts across the *second array dimension* (the time axis, `axis=1`):

$$S_{i,n} = \sum_{t=1}^{1280} s_{i,t,n}. \tag{1}$$

This operation produces an array of shape $(3200, 88)$, with one total spike count per trial and V1 neuron.

We then grouped trials using the stimulus labels supplied in the data. For neuron n and stimulus code θ, the mean response was computed over all trials assigned to that code:

$$\bar{R}_n(\theta) = \frac{1}{N_\theta} \sum_{i:\,\mathrm{stim}_i=\theta} S_{i,n}, \tag{2}$$

where N_θ is the number of repetitions of stimulus θ. Repeating this procedure for every V1 neuron and orientation yields a mean-response matrix of shape $(88, 8)$. In Sections 2–5 these mean spike counts are divided by the stimulus duration of 1.28 s and expressed as firing rates in spikes/s, whereas Figures 1 and 2 show raw mean counts per trial.

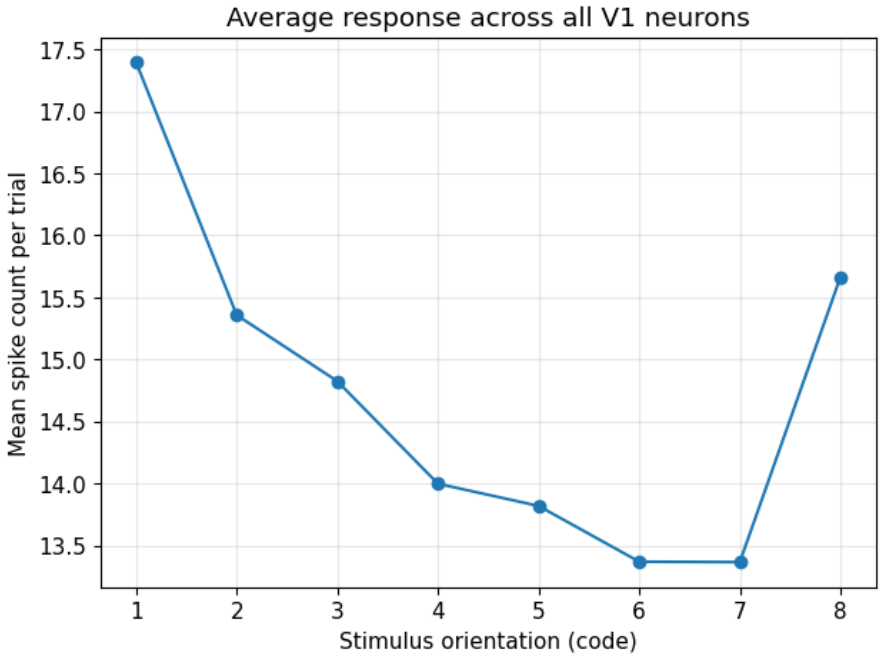


Figure 1: Mean spike count per trial, averaged across all 88 V1 neurons for each stimulus orientation. This is a dataset-level summary; the individual neuron responses were calculated separately.

Questions

1. How many neurons are recorded in the selected session?

Answer. The selected session contains **88 recorded V1 neurons**. The loaded file also includes 24 V2 neurons, but our analysis focuses on V1. The V1 spike-count array has shape $(3200, 1280, 88)$.

2. How many trials were presented for each orientation?

Answer. There are **400 trials per orientation**, for a total of $8 \times 400 = 3200$ trials. This was verified by counting the occurrences of each stimulus code in the `stim` array.

3. What are the stimulus orientations?

Answer. The dataset contains eight stimulus categories, encoded as integers 1 through 8. Based on the dataset documentation (rather than on numerical values stored in the stimulus-code array), the corresponding angles are 0°, 22.5°, 45°, 67.5°, 90°, 112.5°, 135°, and 157.5°. The orientations are therefore spaced by 22.5°. This mapping follows the dataset README.[1]

Note: For the initial polynomial and Gaussian fits, we use stimulus codes 1–8 as the independent variable. These codes are linearly related to angles by $\alpha = 22.5°(\theta - 1)$.

4. Plot mean orientation tuning curves for several neurons.

[1]Dataset: https://doi.org/10.5061/dryad.h9w0vt4s0.

Answer. Figure 2 shows the mean spike count per trial as a function of stimulus code for six V1 neurons. Their response profiles differ markedly: some have clear peaks, while others show weaker or more irregular dependence on orientation.

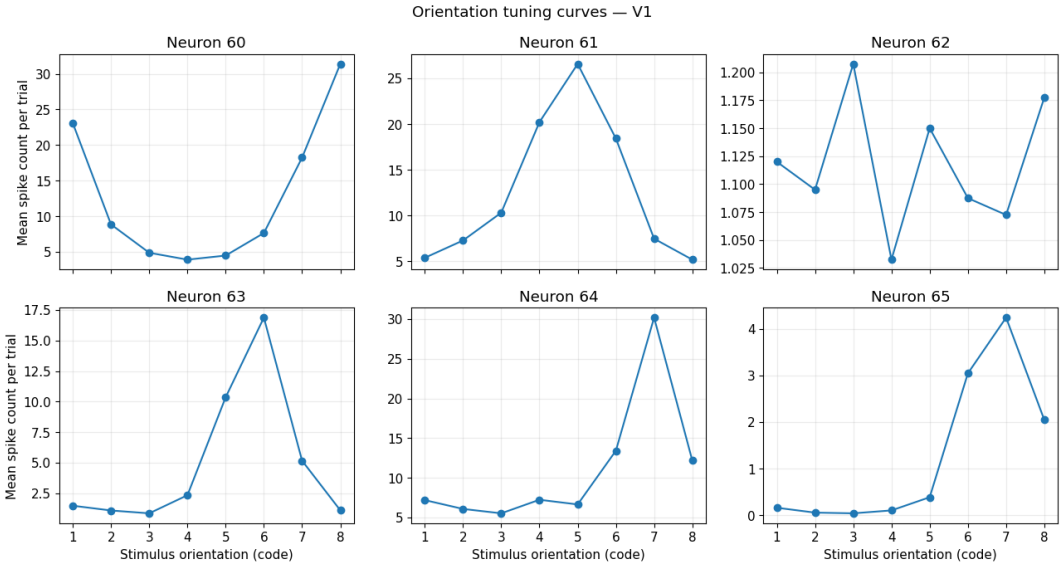


Figure 2: Mean orientation tuning curves for V1 neurons 60–65. Each point is the mean spike count over trials sharing one stimulus orientation.

5. Select one neuron for the remainder of the assignment.

Answer. We selected **V1 neuron 61** (Python index 61). Its tuning curve is approximately bell-shaped, with a pronounced maximum around stimulus code 5 and lower responses toward both ends of the tested range. This relatively clear orientation dependence makes it suitable for comparing polynomial and Gaussian fits. We use this same neuron throughout the subsequent analyses.

2 Polynomial Regression

We fitted four polynomial models to the eight mean firing rates of V1 neuron 61: degrees 1, 3, 4, and 7. Each spike count was divided by the stimulus duration of 1.28 s, so the response variable is in spikes/s. We use stimulus codes 1–8 on the horizontal axis (dimensionless category codes). A degree-d polynomial is defined by

$$\hat{R}(\theta) = \beta_0 + \beta_1\theta + \cdots + \beta_d\theta^d. \tag{3}$$

The goodness of fit is evaluated using the mean squared error (MSE) over the eight stimulus categories. MSE has units $(\text{spikes/s})^2$:

$$\text{MSE} = \frac{1}{8}\sum_{j=1}^{8}\left(\bar{R}_{61}(\theta_j) - \hat{R}(\theta_j)\right)^2. \tag{4}$$

Questions

1. Which polynomial degree provides the best fit?

Answer. The seventh-degree polynomial has the lowest training MSE (reported as 0.0000). With eight distinct input values and eight polynomial coefficients, it can interpolate the eight

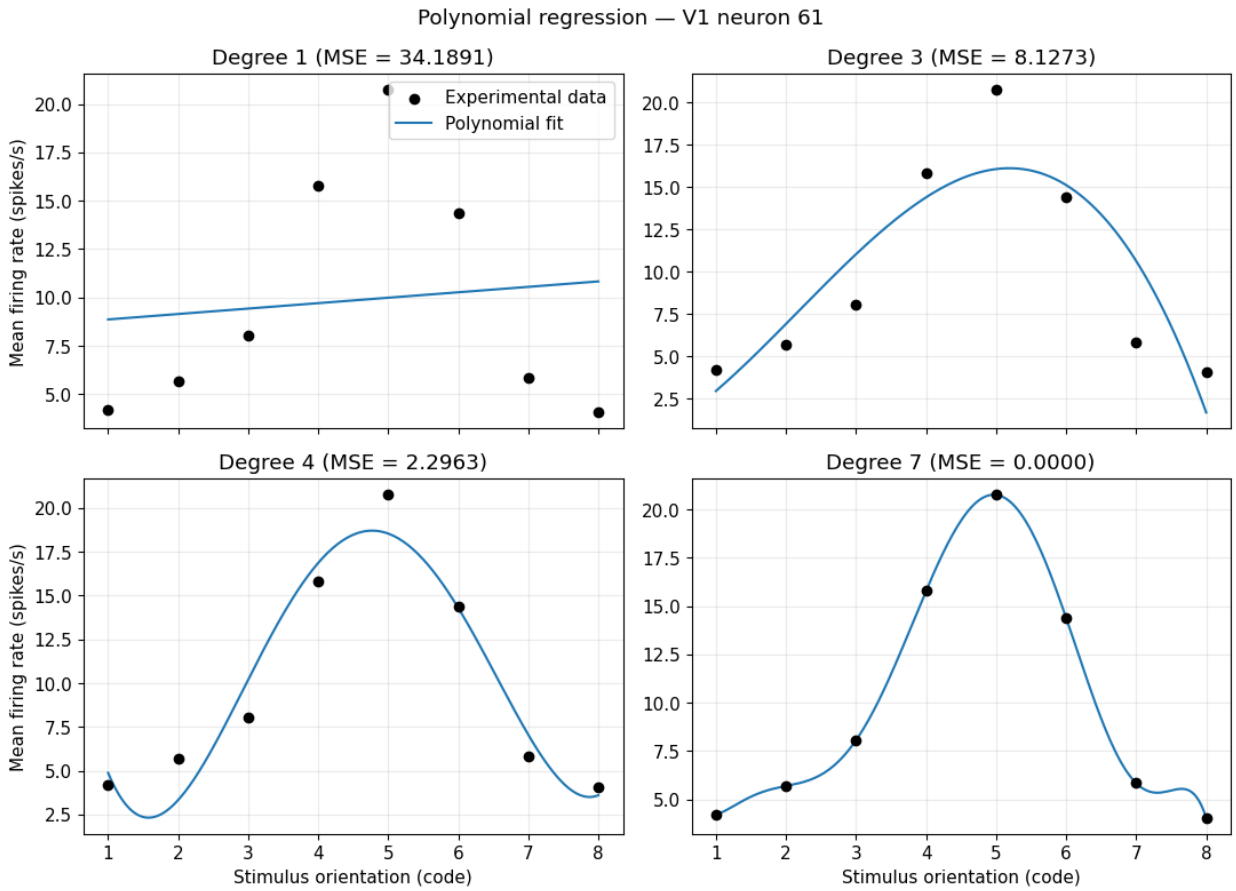


Figure 3: Polynomial regression for neuron 61 at degrees 1, 3, 4 and 7. Black markers show experimental mean firing rates (spikes/s), and blue curves are the fitted models. The x-axis uses orientation codes 1–8. MSE is given in $(\text{spikes/s})^2$.

Table 1: Training MSE in firing-rate units for neuron 61.

Model	Training MSE
Linear (degree 1)	34.1891
Cubic (degree 3)	8.1273
Quartic (degree 4)	2.2963
Degree 7	0.0000

mean-response points exactly, up to numerical precision. This is the best fit *to the available mean responses*, not necessarily the best model for new observations.

2. How does the fit change as the polynomial degree increases?

Answer. Increasing the degree reduces the training MSE from 34.1891 (degree 1) to 8.1273 (degree 3), 2.2963 (degree 4) and essentially 0 (degree 7). The linear fit is almost flat and misses the peak entirely. The cubic fit captures a single hump but underestimates the peak. The quartic fit follows the peak well but dips below the data near codes 1–2. The degree-7 polynomial passes through all eight points, but shows a small oscillation between codes 7 and 8 that has no counterpart in the data, a typical sign of overfitting.

3. What does polynomial degree control?

Answer. Polynomial degree controls the model's complexity and flexibility: a higher degree provides more coefficients and allows more changes in curvature. However, flexibility alone does not guarantee better generalization.

4. What are the advantages and disadvantages of a higher-degree polynomial?

Answer. A higher-degree polynomial can capture complex nonlinear relationships and substantially reduce training error. Its drawbacks include a risk of overfitting, unstable predictions between or beyond measured orientations, and limited interpretability.

5. Can the polynomial coefficients be interpreted biologically?

Answer. Not directly. Individual coefficients are mathematical weights used to construct the fitted curve; they do not separately represent a neuron's preferred orientation, baseline firing or orientation selectivity.

3 Nonlinear Regression: Gaussian Tuning Model

Polynomial regression can reproduce the shape of an orientation tuning curve, but its coefficients have no straightforward biological interpretation. We therefore fit a Gaussian tuning function to the mean responses of V1 neuron 61. The Gaussian describes a neuron whose response peaks at one preferred orientation and decreases as the stimulus moves away from that preference.

$$R(\theta) = B + A \exp\left[-\frac{(\theta - \mu)^2}{2\sigma^2}\right]. \tag{5}$$

Table 2: Parameters of the Gaussian tuning model.

Symbol	Parameter	Interpretation
$R(\theta)$	Predicted response	Mean firing rate (spikes/s) predicted for stimulus orientation θ.
θ	Stimulus orientation	Input orientation (represented initially by code 1–8 in our case, but can be in degrees).
B	Baseline response	Model response far from the preferred orientation.
A	Response amplitude	Difference between peak response and baseline.
μ	Preferred orientation	Location of the model's response peak, assuming $A > 0$.
σ	Tuning width	Width of the Gaussian; reflects how narrowly or broadly the neuron is tuned.

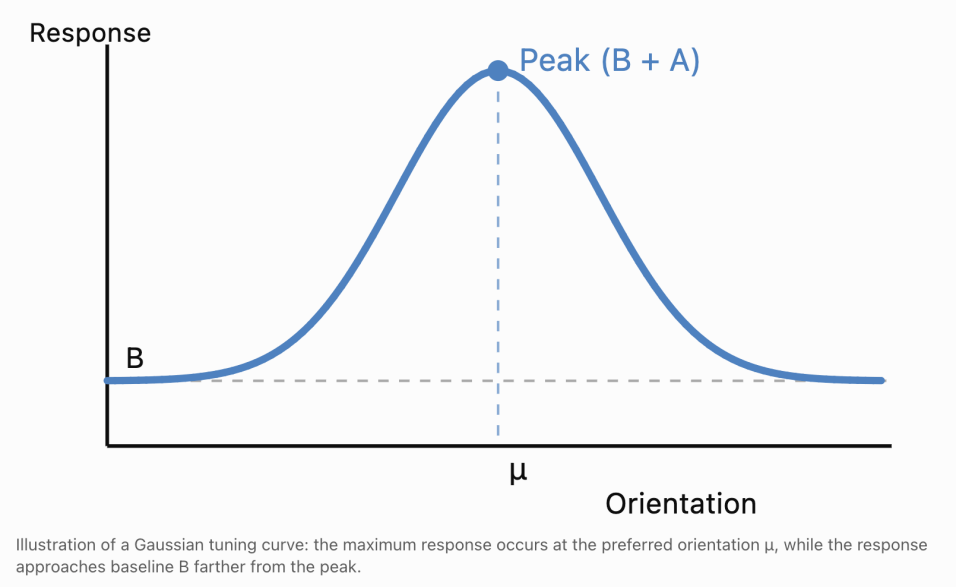


Figure 4: Schematic Gaussian tuning curve, indicating its baseline, preferred orientation, and peak response. Illustration generated with ChatGPT (not computed from data).



Figure 5: Illustrative effects of changing B, A, μ, and σ. Illustration generated with ChatGPT (not computed from data).

At the preferred orientation, $\theta = \mu$, the predicted response is $R(\mu) = B + A$. Larger values of σ produce broader tuning, whereas smaller values produce sharper tuning. Figure 4 illustrates the model, and Figure 5 shows how its four parameters affect the curve.

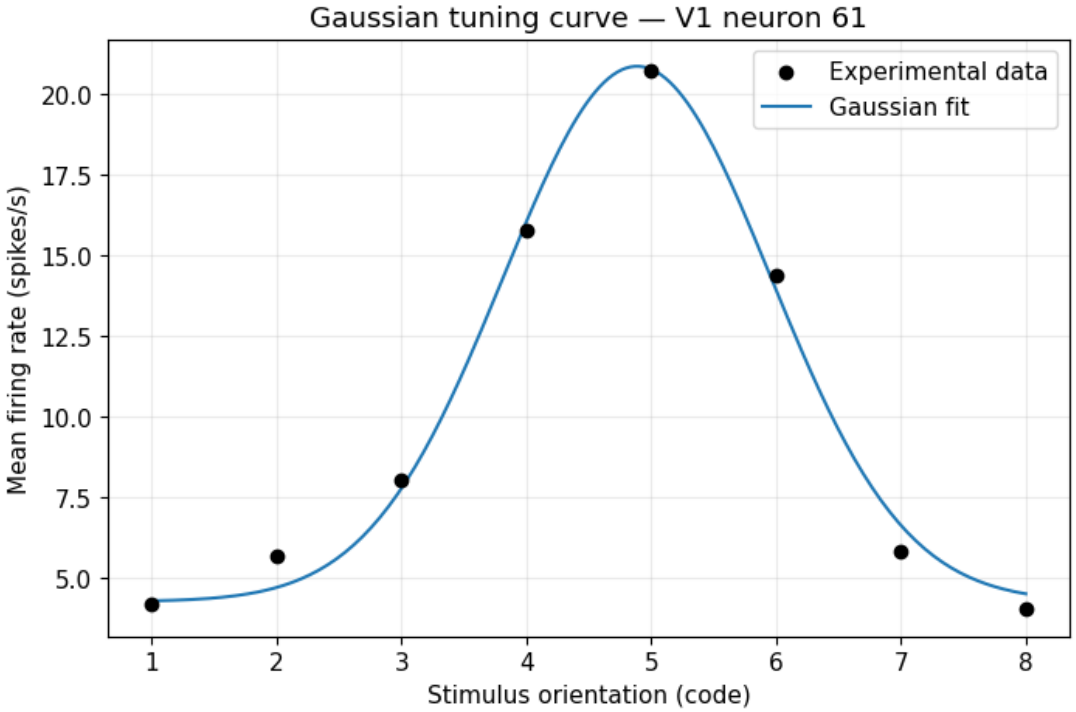


Figure 6: Gaussian model fitted to the mean firing rate of V1 neuron 61. Black dots are experimental orientation-averaged responses (spikes/s); the blue line shows the fitted model.

For neuron 61, the fitted parameters were $B = 4.284$ spikes/s, $A = 16.591$ spikes/s, $\mu = 4.887$ code units, and $\sigma = 1.070$ code units. The model therefore predicts a maximum mean response of $B + A = 20.875$ spikes/s, near stimulus code 5. The estimated width suggests a preference centered near code 5, but the response is spread over several neighboring orientations. The Gaussian fit has a training MSE of 0.2706 $(\text{spikes/s})^2$. Figure 6 shows the fit to the actual data.

Questions

1. Which model has the lowest MSE?

Answer. The 7th-degree polynomial has the lowest training MSE (0.0000), as it can fit all eight experimental points almost exactly. However, this does not necessarily mean that it will perform better on new data. Among the remaining models, the Gaussian achieved the lowest MSE (0.2706).

2. What is the estimated preferred orientation μ?

Answer. For neuron 61, we obtained $\mu = 4.887$, which is close to stimulus code 5. Under the adopted conversion to physical orientation, this corresponds to approximately 87.46°, near 90°. Thus, neuron 61 shows its strongest predicted response near this orientation.

3. What does σ tell us about the neuron's tuning?

Answer. The parameter σ describes the width of the tuning curve and therefore the neuron's orientation selectivity. A smaller σ indicates a narrower range of strong responses and greater selectivity; a larger σ indicates broader tuning. For neuron 61, $\sigma = 1.070$ stimulus-code

units (about 24.1° on the adopted linear angle scale), so the modeled response is spread over neighboring orientations rather than restricted to one code.

4. **Which parameters have a biological interpretation?**

   Answer. All four fitted parameters have interpretable roles. The baseline B represents the response away from the preferred orientation; A measures the additional peak response above baseline; μ estimates the preferred orientation; and σ describes tuning width or orientation selectivity. The parameter B shifts the curve vertically, whereas μ shifts its peak along the orientation axis.

5. **Why might this nonlinear model be preferable to a polynomial even when their MSE values are similar?**

   Answer. The Gaussian model may be preferable because its parameters have a biological meaning. They describe the neuron's baseline response, preferred orientation, and tuning width. A high-degree polynomial may fit the data more closely, but it can also overfit and its coefficients are harder to interpret.

⋆ Supplementary comparison: Gaussian fit to a non-bell-shaped neuron

As an extra check, we also tried fitting the same Gaussian function to V1 neuron 0, whose tuning curve is less bell-shaped. We wanted to see whether the fitted parameters would still be easy to interpret. Neuron 61 remains our main neuron for the assignment; this is only an additional example.

Unlike neuron 61, whose mean responses show a clear peak around orientation code 5, neuron 0 has a less regular orientation-response profile. Its measured mean spike counts before conversion to firing rates were

$$\begin{aligned}\mathbf{y}_0 = (&18.560,\ 16.065,\ 13.4525,\ 14.1425,\\ &16.240,\ 14.8675,\ 13.345,\ 16.655).\end{aligned} \tag{6}$$

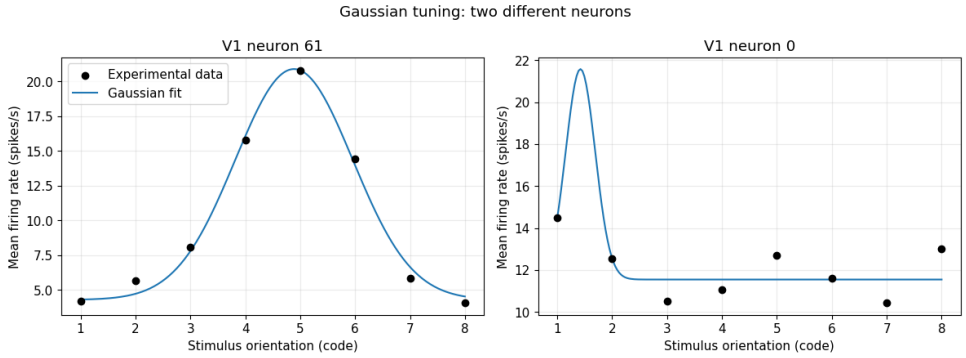


Figure 7: Comparison of Gaussian fits for V1 neuron 61 (left) and neuron 0 (right), from our merged notebook. Markers are measured mean firing rates in spikes/s. The narrow peak for neuron 0 falls between two sampled stimulus codes and is not itself an observed response.

What we noticed. Neuron 61 has a more regular, bell-shaped response. Its fitted parameters are $B = 4.284$, $A = 16.591$, $\mu = 4.887$, and $\sigma = 1.070$. The preferred orientation is close to code 5, matching the peak we saw in the tuning curve. Its predicted maximum is $B + A = 20.875$ spikes/s. The relatively larger σ suggests that the neuron responds noticeably to several nearby orientations, rather than only one very narrow stimulus range.

Table 3: Gaussian parameter comparison between the primary neuron (61) and the supplementary example (0). The errors are measured on the eight mean responses for each neuron.

Parameter	Neuron 61: bell-shaped	Neuron 0: less bell-shaped
Baseline B (spikes/s)	4.284	11.550
Amplitude A (spikes/s)	16.591	10.028
Preferred orientation μ (code)	4.887	1.422
Tuning width σ (code units)	1.070	0.269
Gaussian training MSE $(\text{spikes/s})^2$	0.2706	0.7540

Neuron 0 gave quite different results: $B = 11.550$ spikes/s, $A = 10.028$ spikes/s, $\mu = 1.422$, and $\sigma = 0.269$. Its measured responses do not follow a clear bell shape, yet the Gaussian fit produces a very narrow peak between codes 1 and 2. At this peak, the model predicts $B + A = 21.578$ spikes/s, although the highest measured mean firing rate was only 14.500 spikes/s. We do not have a measurement at code 1.422, so we cannot tell whether this predicted peak is real.

The most noticeable difference between the fits is the width: $\sigma = 1.070$ for neuron 61, compared with $\sigma = 0.269$ for neuron 0. At first, the smaller value for neuron 0 could suggest stronger orientation selectivity. However, looking at the original data, we think this interpretation would be uncertain. The model appears to be fitting a narrow peak between the sampled orientations rather than describing an obvious pattern in the measurements.

Our conclusion. The Gaussian training MSE is lower for neuron 61 (0.2706) than for neuron 0 (0.7540). However, since these neurons have different response scales, this comparison alone is not enough to say which neuron is better described by the model. This small comparison helped us see why it is worth checking the fitted curve, not only the MSE or parameter values. A Gaussian seems suitable for neuron 61 because its tuning curve is reasonably bell-shaped. For neuron 0, the numerical fit may look good at the measured points, but some of its parameter estimates seem less reliable as descriptions of the actual neural response. We would need more orientations or independent trials to check this properly.

Limitation. Both polynomial and Gaussian fits in this section treat stimulus orientation as an ordinary numerical variable. They do not enforce the equivalence between 0° and 180°. This limitation motivates the circular model in the next part of the assignment.

4 A Problem with Orientation: von Mises Model

Just before, we treated stimulus orientation as an ordinary number. However, orientation is circular: 0° and 180° describe the same orientation. We therefore tried the von Mises function, which takes this property into account.

$$R(\theta) = B + A \exp\left[\kappa \cos\left(2(\theta - \mu)\right)\right]. \tag{7}$$

Here, B is an additive baseline parameter, A controls response size, μ is the preferred orientation, and κ controls the concentration around that orientation. We used angles in radians when fitting the model, since the cosine function in Python expects radians. Unlike the Gaussian, this particular parameterization has a peak of $B + Ae^{\kappa}$, rather than $B + A$.

Questions about circularity

1. **Explain why circularity is a problem for the Gaussian model.**

   Answer. A regular Gaussian uses numerical distances between angles. This means it treats 0° and 180° as far apart, even though they describe the same line orientation. It may therefore predict quite different responses to equivalent orientations.

2. **Why is a Gaussian near 0° difficult to use across the boundary?**

   Answer. A Gaussian centered near 0° will tend to fall toward its baseline as the angle approaches 180°. In reality, a neuron preferring nearly horizontal lines might respond strongly on both sides of that boundary. The Gaussian does not automatically join those ends together.

3. **Why is polynomial regression also problematic?**

   Answer. A polynomial also sees the first and last angles as separate ends of a numerical interval. It does not require their predictions to agree and may behave oddly near the ends, particularly for high-degree fits.

4. **How might this affect our selected neuron?**

   Answer. We expected a relatively small effect for neuron 61, because its strongest response is around 90°, well away from the boundary. This seems consistent with our results: both Gaussian and von Mises gave similar tuning curves, although Gaussian had a slightly lower training MSE.

Fitting the circular model

The factor of 2 inside the cosine is important: it makes the function repeat after 180°, rather than 360°. For neuron 61, the fit gave $B = 3.097$, $A = 4.249$, $\mu = 87.420°$, and $\kappa = 1.439$. The training MSE was 0.3051 (spikes/s)^2.

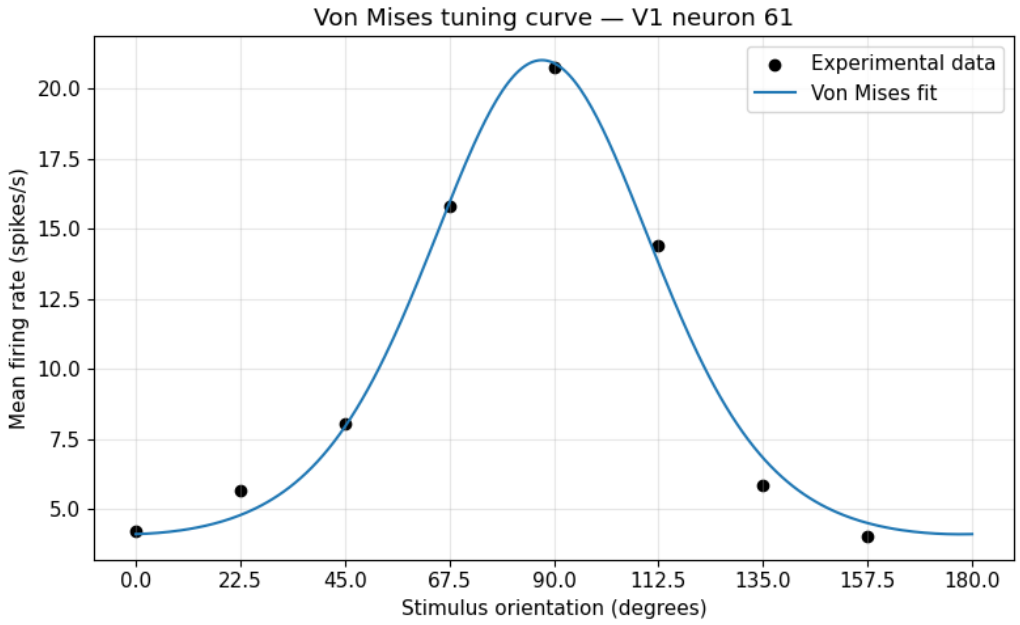


Figure 8: von Mises tuning fit for V1 neuron 61. Black points are experimental mean firing rates and the blue line is the fitted circular model. The predicted values match at 0° and 180°.

For comparison, the Gaussian MSE for neuron 61 was 0.2706, slightly lower than the von Mises value. We also fitted von Mises to our extra example, neuron 0. There, its MSE was 0.5134,

compared with a Gaussian MSE of 0.7540. This extra result suggests that the circular model can be useful for a neuron with a different tuning shape, but it does not establish that von Mises is always better.

Questions about the von Mises model

1. How does the von Mises fit compare with the Gaussian fit?

Answer. Both models fit neuron 61 quite well and estimate a preferred orientation close to 90°. The Gaussian has a slightly lower MSE (0.2706 compared with 0.3051). However, von Mises accounts for the circular nature of orientation and predicts the same response at 0° and 180°, unlike the Gaussian.

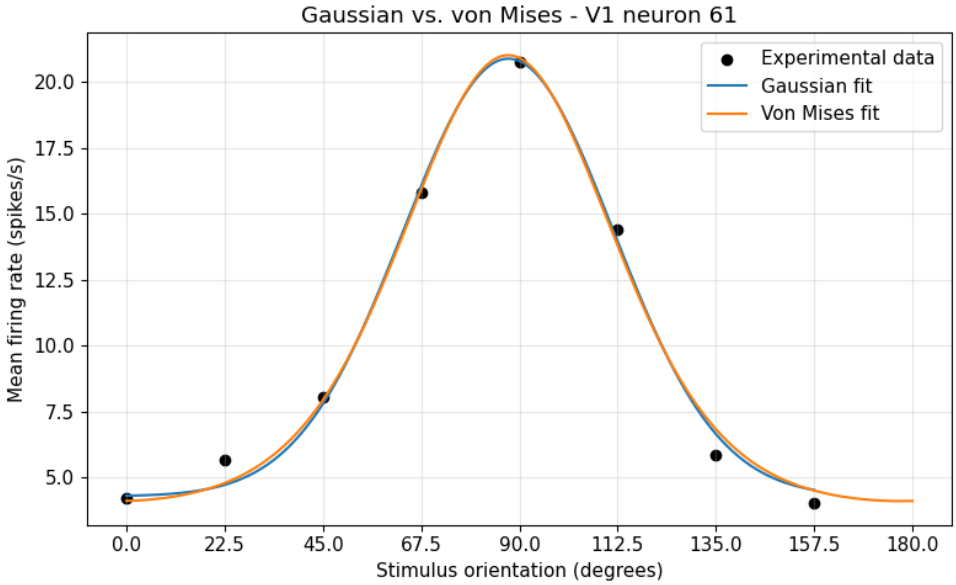


Figure 9: Gaussian and von Mises fits for V1 neuron 61 plotted on the same orientation axis (degrees). Black points are experimental mean firing rates. The two curves are almost indistinguishable.

2. What is the estimated preferred orientation?

Answer. The von Mises fit gives $\mu = 87.42°$, close to the 90° stimulus and to the Gaussian estimate of 87.46°. The two independent estimates agree to within 0.05°, which supports the conclusion that neuron 61 prefers an orientation close to 90°.

3. What does the parameter κ tell you about the neuron's tuning?

Answer. κ controls how sharply the neuron is tuned. A small κ gives a flat curve with almost no orientation preference, and a larger κ gives a narrower and more selective peak. For neuron 61, $\kappa = 1.439$ produces a clear single peak around 87° that falls to a low response at the orthogonal orientation. The width of this peak is similar to the width of the Gaussian fit, so both models describe about the same breadth of tuning.

4. Why does the equation contain $2(\theta - \mu)$ rather than simply $(\theta - \mu)$?

Answer. Orientation has a period of 180°, whereas $\cos(\theta - \mu)$ has a period of 360° and would treat 0° and 180° as opposite points ($\cos 180° = -1$), giving different predictions for physically identical stimuli. With the factor 2, shifting θ by 180° adds 360° inside the cosine, so $R(\theta + 180°) = R(\theta)$ and the curve is automatically periodic with the period of orientation.

5. Does accounting for circularity improve the fit for your selected neuron?

Answer. Not for neuron 61. The von Mises model has a slightly higher MSE (0.3051) than the Gaussian (0.2706). This may be because the neuron's preferred orientation is close to 90°, so the 0°/180° boundary has little influence on its tuning curve.

6. Which model would you choose to describe this neuron's orientation tuning, and why?

Answer. We would choose the von Mises model. Its training MSE is almost the same as the Gaussian (difference ≈ 0.03), it has the same number of parameters, all of them are interpretable (μ, κ, and the amplitude and baseline terms), and it is the only one of the two that respects the periodicity of orientation. The seventh-degree polynomial is excluded because it interpolates the eight means exactly and has no biological interpretation, and the linear, cubic and quartic polynomials fit considerably worse. For this particular neuron, the Gaussian would be an equally acceptable description.

5 Comparing the models and evaluating generalization

5.1 Comparison for V1 neuron 61

Table 4 brings together the six fits we tested on our main neuron. All MSE values are calculated from the eight mean firing rates and expressed in $(\text{spikes/s})^2$.

Table 4: Model comparison for V1 neuron 61. The parameter counts include the intercept or baseline parameter.

Model	Parameters	MSE	Biological meaning?	Circular?
Linear	2	34.1891	No	No
3rd-order polynomial	4	8.1273	No	No
4th-order polynomial	5	2.2963	No	No
7th-order polynomial	8	0.0000	No	No
Gaussian	4	0.2706	Yes	No
von Mises	4	0.3051	Yes	Yes

The seventh-degree polynomial has the lowest training MSE, almost zero, but it uses eight parameters for only eight mean responses. This could be a sign of overfitting, so we would not choose it based only on its training error. The Gaussian and von Mises models both fit neuron 61 fairly well with only four parameters, and their parameters are easier to relate to neuronal tuning. Gaussian is slightly better in terms of training MSE for this particular neuron, while von Mises has the advantage of respecting the circular nature of orientation.

Extra comparison with neuron 0. For neuron 0, the Gaussian MSE was 0.7540 and the von Mises MSE was 0.5134, both in $(\text{spikes/s})^2$. Unlike neuron 61, neuron 0 showed a lower error with von Mises. We think this is worth noting, although we would need additional data to say much about generalization.

5.2 Comparing neurons with different noise levels

For the final comparison, we returned to the trial-level recordings from Part 1. For each neuron we calculated the standard deviation of single-trial spike counts across repeated trials for each orientation and averaged it over the eight orientations. We ranked the 88 V1 neurons by this

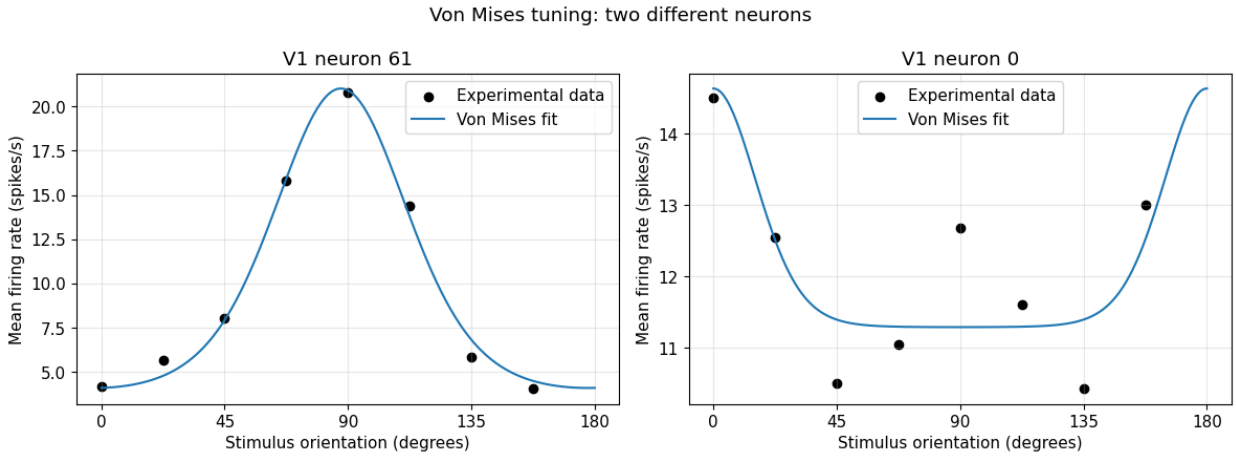


Figure 10: Additional von Mises fits for neurons 61 and 0. Both fitted curves are periodic over 180°, although the two neurons have quite different response patterns.

measure and plotted the tuning curves of the ten most variable ones. From these we selected **neurons 12 and 50**. We treat these two neurons as examples with relatively variable responses.

We fitted the same three models to both neurons: a fourth-order polynomial, a Gaussian, and von Mises. The Gaussian model was required by the assignment; the polynomial provides a more flexible comparison, while von Mises also takes orientation circularity into account, and we calculated training MSE.

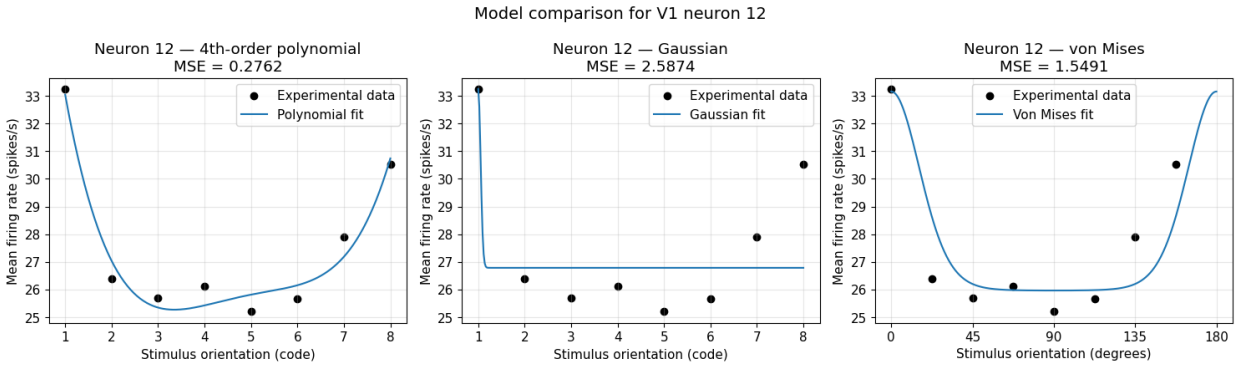


Figure 11: Fourth-order polynomial, Gaussian and von Mises fits for V1 neuron 12. The black points show measured mean firing rates, and the blue lines show the fitted models. Gaussian and polynomial plots use stimulus codes; von Mises uses degrees.

Table 5: Training MSE of the same three models for the selected neuron (61) and the two more variable neurons (12 and 50), in $(\text{spikes/s})^2$.

Model	Neuron 61	Neuron 12	Neuron 50
Fourth-order polynomial	2.2963	0.2762	0.4130
Gaussian	0.2706	2.5874	6.7208
von Mises	0.3051	1.5491	2.8362

Questions

1. Compare the goodness of fit of the different models for the two neurons.

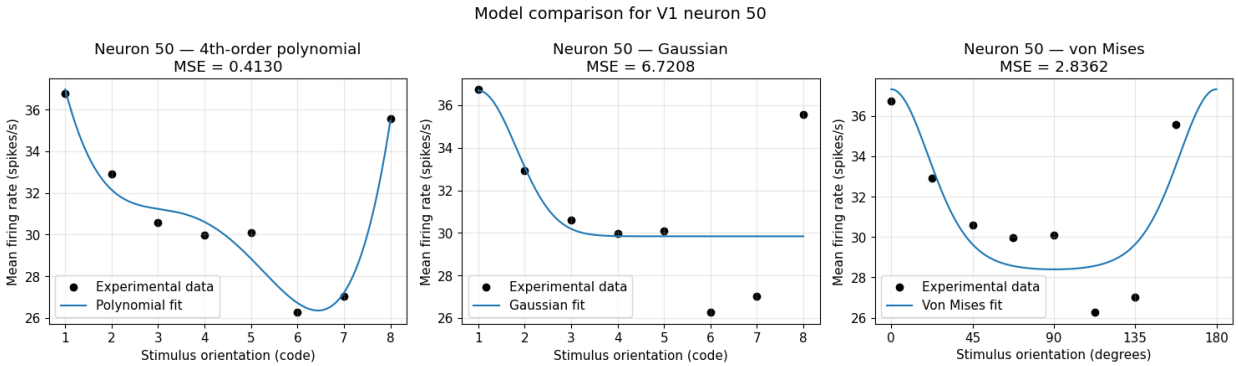


Figure 12: The same three models fitted to V1 neuron 50. The fitted polynomial follows the measured points more closely, while von Mises captures the increased responses near the 0°/180° boundary.

Answer. The fourth-order polynomial has the lowest training MSE for both neurons: 0.2762 for neuron 12 and 0.4130 for neuron 50. Von Mises performs better than Gaussian in both cases, although it does not fit the eight measured means as closely as the polynomial. From the plots, both neurons seem to respond more strongly near the ends of the orientation range, with lower responses around the middle. Von Mises can describe these two ends as part of one circular tuning pattern.

2. How does increased noise affect the relative performance of the models?

Answer. Higher noise may make it harder to identify the neuron's actual tuning pattern. In our results, the 4th-order polynomial fits the observed responses most closely for both neurons 12 and 50. However, it may also follow some random variations in the data. The Gaussian performs worse, possibly because these neurons have a different tuning shape, with stronger responses near the 0°/180° boundary.

3. Does the model with the lowest MSE necessarily generalize best?

Answer. No. The values in Table 5 are training errors calculated on the same eight mean responses that were used for fitting. A flexible model can follow random fluctuations in these means and so reach a low training MSE while predicting new orientations or new trials worse. The fourth-order polynomial has five parameters for eight data points, so its lower error for neurons 12 and 50 is partly explained by its flexibility.

4. Which model would you choose to predict a response to a new stimulus orientation?

Answer. We would choose von Mises. Greater model complexity reduces the training error but increases the risk of fitting noise, and the noisier the data, the more this risk matters. The von Mises model has only four parameters, its shape is constrained to a smooth periodic peak, and it respects the 0°/180° boundary. This is useful for neurons 12 and 50, whose responses are high at both ends of the measured range. Its training MSE is higher than that of the polynomial, but this extra error is the price of a simpler model that is less likely to reproduce noise. A held-out test set would be needed to confirm this.